

MBA979 — Project Module Report

Empirical Asset Pricing via Machine Learning: A Reproduction and Adaptation on Indian Equities (Nifty 50)

Field	Detail
Course	MBA979 — Project Module
Report type	Project Review / Module Report
Base paper	Gu, Kelly, and Xiu (2020), <i>Empirical Asset Pricing via Machine Learning</i> , <i>The Review of Financial Studies</i> , 33(5), 2223–2273
Market studied	India — Nifty 50 equities
Sample period	January 2015 – December 2024
Code repository	https://github.com/balajibrk/gkx-india
Implementation	Python (scikit-learn, LightGBM, PyTorch)

1. Executive Summary

This project reproduces the core empirical framework of Gu, Kelly, and Xiu (2020) — hereafter GKX (2020) — and adapts it to the Indian equity market. Using monthly data on Nifty 50 constituents over 2015–2024, we construct a panel of stock characteristics interacted with India-relevant macroeconomic variables, train nine return-prediction models spanning linear, tree-based, and neural-network families, and evaluate them with out-of-sample (R^2), information coefficients, Diebold–Mariano tests, and long–short portfolio performance.

Main finding. Consistent with GKX (2020), the deepest neural network (NN3) achieves the highest out-of-sample predictive (R^2) on the 2022–2024 test sample (**+1.0458%**), outperforming the OLS-3 baseline (**+0.6458%**). Macro–characteristic interaction features dominate variable importance. Portfolio results are noisier on a 47-stock universe: PCR yields the strongest long–short Sharpe ratio (**0.79**), while NN3’s ranking signal is weaker despite its superior (R^2) — a known small-sample tension between squared-error fit and cross-sectional ranking quality.

All code, metric tables, prediction files, and figures are available in the public repository linked above.

2. Base Paper: Gu, Kelly, and Xiu (2020)

2.1 Bibliographic details

Gu, Shihao, Bryan Kelly, and Dacheng Xiu. 2020. “Empirical Asset Pricing via Machine Learning.” *The Review of Financial Studies* 33 (5): 2223–2273. <https://doi.org/10.1093/rfs/hhaa009>

2.2 Abstract of the base paper (summary)

GKX (2020) study the problem of predicting the cross-section of individual U.S. stock returns using a large set of firm characteristics and their interactions with macroeconomic state variables. They compare a wide range of machine-learning methods — including ordinary least squares with few predictors, penalised linear models (elastic net), dimension-reduction methods (principal component regression and partial least squares), regression trees and boosted trees, and multilayer neural networks — against a common empirical design.

The paper’s central conclusions are:

- 1. **Predictive gains from ML.** Flexible nonlinear methods, especially neural networks and gradient-boosted trees, produce substantially higher out-of-sample (R^2) than simple linear benchmarks.
- 2. **Economic value.** Predictions translate into economically meaningful long–short portfolio returns and Sharpe ratios.
- 3. **Which signals matter.** A relatively small subset of predictors — especially momentum-related characteristics and interactions with macro variables — accounts for most of the predictive content.
- 4. **Regularisation is essential.** Unconstrained high-dimensional OLS fails; shrinkage, dimension reduction, and ensemble methods are necessary to control overfitting.

The original study uses a long U.S. panel (roughly 1957–2016), a very large cross-section of stocks, and dozens of CRSP/Compustat-style characteristics. Our project keeps the *methodological skeleton* of GKX and transplants it to a shorter, smaller Indian sample with India-specific macros.

2.3 What this project reproduces vs. adapts

Dimension	GKX (2020) — U.S.	This project — India
Universe	Broad CRSP equities (~thousands)	Nifty 50 (≈47 stocks with usable history)
Period	Multi-decade (≈1957–2016)	Jan 2015 – Dec 2024
Characteristics	≈94 firm signals	17 price/volume-based characteristics
Macro variables	U.S. macro states	8 India-focused macros
Feature design	Char × macro interactions	Same logic → 153 features
Models	OLS-3, ENet, PCR, PLS, RF, GBRT, NN1–NN5 (paper)	OLS-3, ENet, PCR, PLS, RF, GBRT, NN1–NN3
Portfolio sorts	Deciles	Quintiles (appropriate for ~47 names)
Primary metric	(R^2_{oos}) vs. zero forecast	Same
Evaluation extras	IC, DM tests, portfolios	Same suite

3. Research Question and Objectives

3.1 Research question

Can machine-learning methods improve monthly cross-sectional return prediction for Indian large-cap equities relative to simple linear benchmarks, and does the GKX finding that nonlinear models dominate

linear ones hold on a Nifty 50 sample?

3.2 Project objectives

1. Reconstruct the GKX empirical pipeline (features → train/val/test → model zoo → evaluation → portfolios).
2. Assemble a reproducible Indian dataset with clearly documented sources and sample periods.
3. Train and compare nine models under a common protocol.
4. Report statistical and economic performance honestly, including limitations induced by the small cross-section and short history.
5. Deliver a public, reproducible codebase for academic review.

4. Data

4.1 Overall sample design

Item	Specification
Equity universe	Nifty 50 constituents (NSE), as listed in project config
Stocks successfully used	≈47 (after data-availability filters)
Frequency	Monthly
Calendar coverage	2015-01-01 to 2024-12-31
Train sample	2015-01 → 2019-12
Validation sample	2020-01 → 2021-12
Out-of-sample test	2022-01 → 2024-12
Prediction target	One-month-ahead excess return

The train / validation / test split follows the GKX logic of chronological separation: hyperparameters are chosen on validation; all headline results use the held-out test period only.

4.2 Equity price and return data

Field	Detail
Primary source	Upstox API v2 (NSE equity monthly OHLCV)
Fallback source	Yahoo Finance (yfinance) adjusted closes, if Upstox is unavailable
Period requested	2015-01-01 – 2024-12-31
Instruments	Nifty 50 symbols (e.g. RELIANCE, TCS, HDFCBANK, INFY, ...)
Processed object	Monthly panel of prices → simple returns → excess returns

Excess returns are formed relative to a market/risk-free proxy constructed in the feature pipeline (market return from the cross-section / index side of the build). All characteristics are **lagged by one month** to avoid look-ahead bias.

4.3 Macroeconomic and market-state variables

Eight macro (or market-state) series are aligned to the same monthly grid **2015-01 to 2024-12**:

#	Variable	Description	Source	Identifier / ticker	Period
1	repo_rate	India long-term interest-rate proxy (scaled)	FRED (via pandas-datareader)	INDIRLTLT01STM	2015–2024
2	india_vix	India VIX (volatility / fear index)	Yahoo Finance	^INDIAVIX	2015–2024
3	inr_usd	INR per USD exchange rate	Yahoo Finance	INR=X	2015–2024
4	cpi_yoy	India CPI year-over-year inflation	FRED	INDCPIALLMINMEI (12-month pct change)	2015–2024
5	nifty_pe	Nifty aggregate P/E	Manual NSE CSV if present; else calibrated synthetic series	local / synthetic	2015–2024
6	nifty_pb	Nifty aggregate P/B	Same as above	local / synthetic	2015–2024
7	ep_macro	Aggregate earnings yield	Derived	$(1 / \text{nifty_pe})$	2015–2024
8	bm_macro	Aggregate book-to-market	Derived	$(1 / \text{nifty_pb})$	2015–2024

Macro series are month-end aligned, forward-filled where needed for sparse months, and standardised before interaction with characteristics.

Note for review: If an NSE PE/PB CSV is not supplied, the pipeline falls back to a synthetic PE/PB path (documented in code). Interest-rate and inflation series prefer FRED; FX and India VIX prefer Yahoo Finance. Equity prices prefer Upstox with Yahoo Finance fallback.

4.4 Firm characteristics (17)

Following the spirit of GKX Section 2.1, but restricted to signals computable from the price/volume panel (no full Compustat-style fundamentals):

Group	Characteristics
Momentum / reversal	mom1m, mom3m, mom6m, mom12m, mom36m, reversal
Risk / volatility	beta, retvol, idiovol, maxret
Liquidity / trading	turnover, turnover_vol, dolvol (and related volume features as implemented)
Valuation proxies from prices	size / market-equity style signals where available from the panel

Each characteristic is:

1. Computed with a **one-month lag**.
2. **Cross-sectionally rank-normalised** each month to the interval $[-1, 1]$.

4.5 Final feature matrix

[
 $Z_{\{i,t\}} = \big[c_{\{i,t\}} \otimes x_t \big]$
]

- 17 standalone characteristics ($c_{\{i,t\}}$)
- $17 \times 8 = 136$ interaction terms with macros (x_t)
- **Total: 153 features**

Missing feature values after construction are filled with zeros (post rank-normalisation), preserving panel balance for ML training.

5. Methodology (Step by Step)

This section walks through the full empirical pipeline exactly as implemented in the repository.

Step 1 — Fetch equity prices (`data/01_fetch_prices.py`)

1. Load the Nifty 50 symbol list from `config/settings.py`.
2. For each symbol, request monthly OHLCV from Upstox API v2 for 2015–2024.
3. Resolve instrument keys via ISIN search when symbol-only keys fail.
4. If Upstox fails for a name, fall back to Yahoo Finance.
5. Cache per-symbol files under `data/raw/` (local only; not published) and build a combined returns panel.

Step 2 — Fetch macro panel (`data/02_fetch_macro.py`)

1. Download FRED series for India rates and CPI.
2. Download Yahoo Finance series for India VIX and INR/USD.
3. Load or synthesise Nifty PE/PB; construct `ep_macro` and `bm_macro`.
4. Align all series to a common monthly index covering 2015–2024.
5. Save `macro_panel.parquet`.

Step 3 — Build features and labels (`data/03_build_features.py`)

1. Compute 17 lagged characteristics from the returns panel.
2. Rank-normalise characteristics cross-sectionally to $([-1,1])$.
3. Standardise macros and form $\text{char} \times \text{macro}$ interactions.
4. Define the prediction label ($y_{\{i,t+1\}}$) = next-month excess return.
5. Assign chronological splits: train / val / test using the dates in Section 4.1.
6. Save `x_features.parquet`, `y_labels.parquet`, `meta.parquet`, and `feature_names.csv`.

Step 4 — Linear models (`models/04_linear_models.py`)

Train four linear benchmarks; tune regularisation / components on the **validation** set; save test predictions.

Model	Description
OLS-3	Sparse OLS baseline with three economically motivated predictors (GKX-style simple benchmark)
Elastic Net	Penalised linear model ($L1 + L2$); (α) and (ℓ_1) -ratio tuned on validation
PCR	Principal component regression; number of components tuned on validation
PLS	Partial least squares; number of components tuned on validation

Step 5 — Tree ensembles (`models/05_tree_models.py`)

Model	Description
Random Forest	Bagged regression trees; depth / feature subsample hyperparameters tuned on validation
GBRT (LightGBM)	Gradient-boosted trees; learning rate, leaves, and related hyperparameters tuned on validation

Feature importances from RF and GBRT are saved and later combined into a rank-based importance table.

Step 6 — Neural networks (`models/06_neural_networks.py`)

Feed-forward networks following the GKX architecture family:

Model	Hidden layers (width)
NN1	32
NN2	32 → 16
NN3	32 → 16 → 8

Training protocol:

- Activations: ReLU
- Batch normalisation and dropout
- Adam optimiser with learning-rate decay
- L1 penalty on weights
- Gradient clipping
- Early stopping on validation loss
- Dropout tuned on validation (especially for NN2)
- **10 random seeds** per architecture; ensemble average of predictions; divergent seeds filtered by validation loss

Training was executed on CPU in this environment (CUDA wheels were unavailable for the installed Python version). Results are numerically valid; only wall-clock time differs from a GPU run.

Step 7 — Statistical evaluation (`eval/07_metrics.py`)

For every model on the test set we compute:

1. **Out-of-sample (R^2)** against a zero forecast (GKX definition):

$$R^2_{\text{oos}}$$

$$= 1 - \frac{\sum (y - \hat{y})^2}{\sum y^2}$$

2. **Information Coefficient (IC):** monthly Spearman rank correlation between predictions and realised returns; report mean IC, (t)-stat, and fraction of months with IC > 0.
3. **Diebold–Mariano (DM) test** of predictive accuracy versus OLS-3.
4. **Variable importance:** combined RF + GBRT ranks.

Step 8 — Portfolio construction (eval/08_portfolio.py)

Each month in the test sample:

1. Rank stocks by model predicted return.
2. Form **quintile** portfolios (Q1 = lowest predicted, Q5 = highest).
3. Construct a long–short spread (long Q5, short Q1), value-weighted when market-cap information is available, otherwise equal-weighted.
4. Compute Sharpe ratio (annualised with $(\sqrt{12})$), annualised mean return, cumulative return, win rate, and maximum drawdown.
5. Compare against a Nifty buy-and-hold style benchmark and estimate a simple factor-adjusted alpha where feasible.

Step 9 — Visualisation and reporting (viz/09_charts.py, notebook)

Publication-style charts:

1. (R^2_{oos}) bar chart
2. Variable importance (top predictors)
3. Validation vs test (R^2) scatter (overfitting diagnosis)
4. Quintile return pattern
5. Cumulative long–short PnL
6. India macro-interaction story

Master narrative notebook: notebooks/00_master_results.ipynb.

6. Results

All figures below refer to the **out-of-sample test window: 2022–2024** unless noted. Validation metrics are used only for model selection / diagnosis.

6.1 Out-of-sample predictive (R^2)

Model	Family	Val (R^2) (%)	Test (R^2) (%)
OLS-3	Linear	1.5564	0.6458
Elastic Net	Linear	2.1609	0.4356
PCR	Linear	1.8217	0.4729
PLS	Linear	2.3478	0.0351
Random Forest	Tree	2.1393	0.0825
GBRT	Tree	1.0206	−0.4166
NN1	Neural	3.9555	−1.4943
NN2	Neural	4.4926	−0.5515
NN3	Neural	3.6290	+1.0458

Interpretation

- NN3 is the best test predictor by (R^2_{oos}), confirming the GKX qualitative result that deeper nonlinear models can beat linear baselines.
- Several flexible models (NN1, NN2, GBRT) show **validation strength but test weakness**, illustrating overfitting risk on a short Indian sample — exactly why GKX emphasise validation-based regularisation and held-out testing.
- OLS-3 remains a surprisingly competitive linear benchmark on this small universe.

6.2 Information coefficients (test period)

Model	Mean IC	(t)-stat	% months IC > 0	Months
OLS-3	0.0092	0.26	60.0%	35
ENet	0.0107	0.35	51.4%	35
PCR	0.0058	0.15	48.6%	35
PLS	−0.0055	−0.17	45.7%	35
RF	0.0022	0.08	48.6%	35
GBRT	0.0078	0.26	45.7%	35
NN1	−0.0365	−1.25	45.7%	35
NN2	−0.0211	−0.79	37.1%	35
NN3	−0.0577	−1.88	31.4%	35

IC magnitudes are small and mostly insignificant — expected with only **35 test months** and ≈ 47 names. Notably, NN3's mean IC is negative even though its (R^2_{oos}) is best. This can occur when squared-error fit improves while cross-sectional rank ordering does not; it is an important caveat for portfolio translation on small samples.

6.3 Diebold–Mariano tests versus OLS-3

Model	DM statistic	(p)-value	Significant at 5%?
ENet	−0.31	0.75	No
PCR	−0.28	0.78	No
PLS	−0.75	0.45	No
RF	−0.45	0.65	No
GBRT	−0.49	0.62	No
NN1	−1.02	0.31	No
NN2	−0.45	0.65	No
NN3	+0.33	0.74	No

No model significantly beats OLS-3 on the DM test at conventional levels. With 35 months, the test is underpowered relative to the multi-decade GKX sample. We therefore treat (R^2_{oos}) ordering and economic portfolio diagnostics as complementary evidence, not as DM-rejected superiority.

6.4 Variable importance (India-specific pattern)

Combined RF + GBRT ranks show that **macro-interacted features dominate** the top of the importance list. Leading examples include:

1. mom12m × india_vix
2. beta × repo_rate
3. retvol × ep_macro / standalone retvol
4. maxret × inr_usd
5. mom12m × ep_macro, mom12m × inr_usd, mom12m × cpi_yoy, mom12m × repo_rate

Economic reading. Indian large-cap returns in this sample appear tightly linked to the **policy-rate / inflation / FX / fear** state of the market. Momentum’s predictive content is state-dependent — amplified when interacted with India VIX and RBI-related rate conditions — which is a natural emerging-market counterpart to GKX’s emphasis on characteristic–macro interactions.

6.5 Long–short quintile portfolio performance (test: 2022–2024)

Model	Sharpe (ann.)	Ann. return	Cumulative return	Win rate	Max drawdown
PCR	0.790	+11.67%	+36.00%	60.0%	−15.46%
RF	0.557	+7.53%	+21.32%	57.1%	−16.67%
GBRT	0.123	+1.62%	+2.18%	51.4%	−20.33%
ENet	−0.198	−2.14%	−7.64%	48.6%	−18.28%
PLS	−0.227	−3.57%	−13.11%	48.6%	−37.00%
NN1	−0.228	−3.96%	−14.56%	34.3%	−30.77%
OLS-3	−0.286	−4.06%	−13.78%	48.6%	−26.11%
NN3	−0.556	−8.06%	−23.40%	42.9%	−28.99%
NN2	−0.675	−9.31%	−25.97%	40.0%	−35.24%

Interpretation

- **PCR** produces the strongest tradable long–short record in this small-universe setting.
- **NN3** does *not* win on Sharpe despite winning on (R^2_{oos}) , consistent with its weak/negative IC.
- For academic scoring of the GKX *prediction* claim, (R^2_{oos}) is the primary metric (as in the paper). Portfolio results are reported transparently as a secondary economic lens and are more fragile when (N_{stocks}) is small.

6.6 Mapping results back to GKX claims

GKX claim	Evidence in this India project	Verdict
Nonlinear ML can beat linear (R^2_{oos})	NN3 test $(R^2) > \text{OLS-3}$	Supported
Regularisation / validation matter	NN1/NN2 strong in val, weak in test	Supported
Macro interactions are important	Top importances are mostly interactions	Supported
ML portfolios dominate economically	PCR strongest; NN3 weak on Sharpe/IC	Partially supported / sample-limited
DM significance vs OLS	No model significant at 5%	Not confirmed (short test window)

7. Discussion

7.1 What transferred cleanly from the U.S. paper

The GKX pipeline — lagged characteristics, macro interactions, chronological validation, a multi-family model zoo, and (R^2_{oos}) versus a zero benchmark — is portable. Even with a fraction of the original data scale, the deepest network still posts the best test (R^2) , and macro-conditioned signals dominate importance rankings.

7.2 What is India-specific

Unlike a U.S. all-equity CRSP universe, Nifty 50 is a **large-cap, high-visibility** index. Macro factors (policy rates, inflation, INR, India VIX) play an outsized role. The prominence of $\text{mom12m} \times \text{india_vix}$ and $\text{beta} \times \text{repo_rate}$ is consistent with a market where global risk appetite and RBI cycles mediate stock-level predictability.

7.3 Why (R^2) and portfolio ranks can disagree

(R^2_{oos}) rewards squared-error accuracy relative to a zero forecast. Quintile long–short performance rewards **stable cross-sectional ranking**. With ≈ 47 stocks and 35 months, a model can shrink overall error (good (R^2)) without sorting winners and losers reliably (weak IC / Sharpe). This project reports both, rather than forcing a single narrative.

8. Limitations

1. **Universe size.** ≈ 47 stocks vs. thousands in GKX $\rightarrow$ noisy IC and portfolio inference.
2. **History length.** 10 years vs. multi-decade U.S. sample $\rightarrow$ DM tests underpowered.
3. **Characteristic breadth.** 17 price/volume signals vs. ≈ 94 in the paper $\rightarrow$ no full fundamental set (earnings quality, accruals, analyst data, etc.).
4. **PE/PB macro path.** May rely on synthetic fallback if NSE fundamentals CSV is absent.
5. **Transaction costs / shorting.** Long-short spreads are theoretical; NSE shorting frictions and costs are not modelled.
6. **Compute environment.** Neural nets trained on CPU in this run; architecture and seeds follow the paper's spirit, but hardware differs from a CUDA production setup.

These limitations are material and should be read as scope constraints of an MBA project reproduction, not as failures of the original paper.

9. Reproducibility and Code Access

Item	Location
Public GitHub repository	https://github.com/balajibrk/gkx-india
Configuration / dates / tickers	config/settings.py
Data scripts	data/01_fetch_prices.py, 02_fetch_macro.py, 03_build_features.py
Models	models/04_linear_models.py, 05_tree_models.py, 06_neural_networks.py
Evaluation	eval/07_metrics.py, 08_portfolio.py
Charts	viz/09_charts.py, viz/figures/
Results tables	results/*.csv
Narrative notebook	notebooks/00_master_results.ipynb
Dependencies	requirements.txt

Recommended run order

```
python data/01_fetch_prices.py
python data/02_fetch_macro.py
python data/03_build_features.py
python models/04_linear_models.py
python models/05_tree_models.py
python models/06_neural_networks.py
python eval/07_metrics.py
python eval/08_portfolio.py
python viz/09_charts.py
```

Secrets (API tokens) are loaded from a local `.env` file and are **not** stored in the public repository. An `.env.example` template is provided.

10. Conclusion

This MBA979 project module reproduces the empirical machine-learning asset-pricing framework of Gu, Kelly, and Xiu (2020) on Indian Nifty 50 data for **January 2015 – December 2024**. Using 153 characteristic and macro-interaction features and nine models, we find that:

- 1. **NN3 attains the best out-of-sample (R^2) (+1.0458%)**, exceeding the OLS-3 baseline — in line with the paper’s nonlinear-advantage claim.
- 2. **Macro-interacted predictors dominate variable importance**, highlighting an India-specific, state-dependent return structure.
- 3. **Portfolio evidence is mixed**: PCR delivers the strongest long–short Sharpe (0.79), while NN3’s ranking metrics are weak, underscoring small-sample limits when moving from statistical fit to implementable sorts.
- 4. The full pipeline, results, and documentation are openly available at <https://github.com/balajibrk/gkx-india> for faculty review and verification.

Overall, the project demonstrates that the GKX methodology is implementable and informative on Indian large-cap equities, while also clarifying which of the paper’s conclusions survive — and which become fragile — under a shorter emerging-market sample.

References

- 1. Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223–2273. <https://doi.org/10.1093/rfs/hhaa009>
- 2. Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- 3. Federal Reserve Bank of St. Louis (FRED). India interest-rate and CPI series (INDIRLTLT01STM, INDCPIALLMINMEI).
- 4. National Stock Exchange of India / Yahoo Finance. Nifty-related market data and India VIX.
- 5. Upstox API v2. NSE equity historical market data.

```
@article{gu2020empirical,
  title   = {Empirical Asset Pricing via Machine Learning},
  author  = {Gu, Shihao and Kelly, Bryan and Xiu, Dacheng},
  journal = {The Review of Financial Studies},
  volume  = {33},
  number  = {5},
  pages   = {2223--2273},
  year    = {2020},
  doi     = {10.1093/rfs/hhaa009}
}
```

Appendix A — Repository structure (for review)

```
gkx_india/
├─ README.md
├─ requirements.txt
├─ config/settings.py
├─ data/01_fetch_prices.py
├─ data/02_fetch_macro.py
├─ data/03_build_features.py
├─ models/04_linear_models.py
├─ models/05_tree_models.py
├─ models/06_neural_networks.py
├─ eval/07_metrics.py
├─ eval/08_portfolio.py
├─ viz/09_charts.py
├─ viz/figures/           # six result charts
├─ results/              # predictions + metric CSVs
├─ notebooks/00_master_results.ipynb
└─ docs/MBA979_Project_Module_Report.md  # this report
```

Appendix B — Data source & period checklist

Dataset	Source	Period used
Nifty 50 monthly OHLCV	Upstox API v2 (fallback: Yahoo Finance)	2015-01 – 2024-12
India long-term rate proxy	FRED INDIRLTLT01STM	2015-01 – 2024-12
India CPI	FRED INDCPIALLMINMEI	2015-01 – 2024-12
India VIX	Yahoo Finance ^INDIAVIX	2015-01 – 2024-12
INR/USD	Yahoo Finance INR=X	2015-01 – 2024-12
Nifty PE / PB	NSE CSV or documented synthetic fallback	2015-01 – 2024-12
Train / Val / Test labels	Constructed in feature pipeline	Train 2015–2019; Val 2020–2021; Test 2022–2024

End of MBA979 Project Module Report